NPTEL Online Certification Course
Fundamentals of Artificial Intelligence
MOOCs; July - Dec 2026
Indian Institute of Technology Guwahati
Assignment No. 4: Detailed Solution

10 Marks

Each question carries 01 Mark each.

Q1. Which of the following statements correctly distinguishes Symbolic AI from Connectionist AI?

A. Symbolic AI is inspired primarily by the structure of the human brain.
B. Symbolic AI represents and manipulates knowledge using symbols, rules and structured representations.
C. Connectionist AI requires all knowledge to be represented explicitly as logical rules.
D. Symbolic AI cannot provide interpretable representations.
Correct Answer: (B)

Q2. Assertion A: A knowledge representation is fundamentally a surrogate.
Reason R: It acts as a substitute for the thing itself, enabling an agent to determine consequences by reasoning about the represented world.

Choose the correct answer:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true, but R is false.
D. A is false, but R is true.

Correct Answer: (A)

Q3. Assertion A: Propositional logic has limited expressive power for representing structured knowledge.
Reason R: Propositional logic cannot directly represent properties of individuals, relations between individuals, or generalizations involving them.

Choose the correct answer:

A. Both A and R are true, and R is the correct explanation of A.
B. Both A and R are true, but R is not the correct explanation of A.
C. A is true, but R is false.
D. A is false, but R is true.

Correct Answer: (A)

Q4. Which of the following is logically equivalent to $\neg(P \rightarrow Q)$?

A. $\neg P \vee Q$
B. $P \wedge \neg Q$

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C. $\neg P \wedge Q$
D. $P \vee \neg Q$

Correct Answer: (B)

Q5. Which statement correctly distinguishes a *predicate* from a *function* in First-Order Logic?

A. Both always evaluate to truth values.
B. A predicate evaluates to an object, whereas a function evaluates to a truth value.
C. A predicate applied to arguments produces a proposition, whereas a function applied to arguments denotes an object.
D. Functions can have arguments, but predicates cannot.
Correct Answer: (C)

For Q6- Q10

Consider the following predicates:

- $\text{Robot}(x)$: x is a robot.
- $\text{Autonomous}(x)$: x is autonomous.
- $\text{Mobile}(x)$: x is mobile.

Q6. Which expression represents: All robots are autonomous.

A. $\forall x(\text{Robot}(x) \wedge \text{Autonomous}(x))$
B. $\forall x(\text{Robot}(x) \rightarrow \text{Autonomous}(x))$
C. $\exists x(\text{Robot}(x) \rightarrow \text{Autonomous}(x))$
D. $\exists x(\text{Robot}(x) \wedge \text{Autonomous}(x))$
Correct Answer: (B)

Q7. Which expression represents: Some robots are mobile.

A. $\forall x(\text{Robot}(x) \rightarrow \text{Mobile}(x))$
B. $\exists x(\text{Robot}(x) \wedge \text{Mobile}(x))$
C. $\forall x(\text{Robot}(x) \wedge \text{Mobile}(x))$
D. $\neg\exists x(\text{Robot}(x) \wedge \text{Mobile}(x))$
Correct Answer: (B)

Q8. Which expression represents: No robot is mobile.

A. $\forall x(\text{Robot}(x) \rightarrow \text{Mobile}(x))$
B. $\exists x(\text{Robot}(x) \wedge \neg\, \text{Mobile}(x))$
C. $\neg\exists x(\text{Robot}(x) \wedge \text{Mobile}(x))$
D. $\exists x(\neg\, \text{Robot}(x) \wedge \text{Mobile}(x))$
Correct Answer: (C)

Q9. The negation of $\forall x(\text{Robot}(x) \rightarrow \text{Autonomous}(x))$ is logically equivalent to:

A. $\forall x(\text{Robot}(x) \wedge \neg\, \text{Autonomous}(x))$
B. $\exists x(\text{Robot}(x) \wedge \neg\, \text{Autonomous}(x))$

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C. $\exists x(\neg \text{Robot}(x) \wedge \text{Autonomous}(x))$

D. $\forall x(\text{Robot}(x) \rightarrow \neg\, \text{Autonomous}(x))$

Correct Answer: (B)

Q10. Suppose $\forall x(\text{Robot}(x) \rightarrow \text{Autonomous}(x))$ and $\forall x(\text{Autonomous}(x) \rightarrow \text{Mobile}(x))$ are true. Which conclusion necessarily follows?

A. $\forall x(\text{Robot}(x) \rightarrow \text{Mobile}(x))$

B. $\forall x(\text{Mobile}(x) \rightarrow \text{Robot}(x))$

C. $\exists x(\text{Robot}(x) \wedge \text{Mobile}(x))$

D. $\forall x(\text{Robot}(x) \rightarrow \neg \text{Mobile}(x))$

Correct Answer: (A)

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